Study Guide, Exercises, and Sample Questions

CSE 4615: Wireless Networks

Islamic University of Technology (IUT)

Lecture 4: Understanding MAC for Managing Shared Wireless Channels

1. Topics Covered

- Wireless vs. wired channel access: shared medium, broadcast nature, and interference.
- CSMA (Carrier Sense Multiple Access): multiple access, carrier sensing, and its limitations in wireless.
- Collision detection in wired networks (CSMA-CD): listen-while-transmit, "fate sharing" of the link.
- Why collision detection fails in wireless networks: signal attenuation, "no fate sharing," hidden terminal problem, and fading.
- Collision avoidance (CSMA-CA): waiting for a free medium and random backoff instead of detection.
- Link-layer acknowledgements (ACK) as a substitute for collision detection.
- CSMA with Collision Avoidance using control frames: RTS (Request to Send), CTS (Clear to Send), DATA, and ACK.
- Virtual carrier sense provided by RTS/CTS and its role in protecting against the hidden terminal problem.
- The medium access problem: dividing the shared channel and scheduling its use.
- Provisioned (scheduled) MAC protocols:
  - TDMA (Time Division Multiple Access): fixed time slots per user, wasted idle slots, out-of-band signaling.
  - FDMA (Frequency Division Multiple Access): fixed frequency bands per user, guard bands, wasted idle bands.
  - CDMA (Code Division Multiple Access): all users share the same frequency and time; simultaneous transmission without interference.
- Trade-offs of provisioned protocols: guaranteed access vs. wasted resources and overhead of coordination.
- Using a single shared wideband channel: robustness to multi-path fading, dynamic access, spatial reuse.
- Concurrency vs. taking turns: spatial reuse for far-apart links, taking turns for nearby links.
- When carrier sense (CS) works well: far-apart pairs vs. nearby pairs.

- Hidden terminal problem: a node cannot hear a distant transmitter, causing undetected collisions at the receiver.
- Exposed terminal problem: a node unnecessarily defers transmission because it hears a sender whose receiver is out of range.
- RTS/CTS mechanism in detail: deference upon hearing CTS (solves hidden terminal); deference upon hearing RTS only (exposed terminal can still transmit).
- Key insight: collisions are spatially located at the receiver, not the transmitter.

2. Focus Areas

- Why wireless is a shared medium and how this fundamentally differs from wired point-to-point links.
- CSMA: definition of "carrier sense" and "multiple access"; why sensing is not always possible in wireless.
- CSMA-CD in wired Ethernet: the listen-while-transmit mechanism and why "fate sharing" makes it feasible.
- Why collision detection is infeasible in wireless: signal strength asymmetry between transmitted and received signals ("no fate sharing"), fading, and the hidden terminal problem.
- CSMA-CA: the rationale for collision avoidance over collision detection in 802.11.
- Role of ACK in wireless MAC: absence of ACK as a signal of packet loss.
- RTS/CTS: purpose of each control frame, the four-step exchange (RTS → CTS → DATA → ACK), and how "virtual carrier sense" prevents hidden terminal collisions.
- Difference between the hidden terminal and exposed terminal problems: causes, consequences, and solutions.
- TDMA: structure of slot rounds, examples of unused slots, and the need for out-of-band coordination.
- FDMA: frequency band allocation, guard bands, and wasted spectrum.
- CDMA: simultaneous same-frequency transmission; distinction from TDMA and FDMA.
- Advantages and disadvantages of TDMA/FDMA vs. contention-based protocols.
- Spatial reuse: why far-apart links can transmit concurrently while nearby links must take turns.
- Thresholded correlation value in carrier sense to determine medium occupancy.
- The key insight: collision location is at the receiver, which motivates receiver-based collision control (RTS/CTS).

3. How to Study These Topics

1. Draw a diagram contrasting wired and wireless channel access: show why independent conversations in wired networks become shared-medium interference in wireless.
2. Step through the CSMA-CD procedure for a wired network; then annotate each step explaining why it breaks down in wireless.
3. Sketch the signal-strength-vs.-space plots for both wired ("fate sharing") and wireless ("no fate sharing") scenarios; use them to explain why collision detection is feasible in one and not the other.
4. Practice drawing the hidden terminal and exposed terminal topologies (A-B-C linear layout); trace what each node hears and what it does under plain CSMA vs. under RTS/CTS.
5. Draw the complete RTS/CTS four-step timeline; label each frame, who sends it, who hears it, and who defers.
6. Create a comparison table for TDMA, FDMA, and CDMA covering: dimension divided, guaranteed access, wasted resource, coordination overhead, and robustness to fading.
7. Sketch a TDMA round (six slots, only stations 1, 3, and 4 active) and an FDMA diagram (six bands, only three in use); explain the efficiency loss in each.
8. Summarize the trade-offs of provisioned vs. contention-based MAC in two or three sentences each.
9. For the spatial reuse concept, draw two topologies (far-apart pairs and nearby pairs) and explain when concurrent transmission is safe and when it is not.
10. Prepare one-page handwritten summaries of:
    - The reasons collision detection fails in wireless and how CSMA-CA addresses them.
    - Hidden terminal vs. exposed terminal: definitions, diagrams, and solutions.
    - TDMA, FDMA, and CDMA side-by-side with advantages and disadvantages.

4. Exercises

4.1. Shared Medium and Carrier Sense

1. Explain why two conversations are independent in a wired network but interfere in a wireless network. Use a diagram in your answer.
2. Define "carrier sense" and "multiple access" in the context of CSMA. Why might a wireless station fail to detect an ongoing transmission even with carrier sense?
3. Describe the two problems that prevent 802.11 from performing collision detection. For each problem, explain the underlying physical cause.
4. Compare CSMA-CD and CSMA-CA: procedure, hardware requirement, efficiency, and typical deployment environment.

4.2. Collision Detection and Fate Sharing

5. Explain the concept of "fate sharing of the link" in a wired network. How does it make collision detection feasible at the transmitter?

6. Draw the signal-strength-vs.-space diagram for a wired network (nodes A, B, C). Identify where a collision occurs and explain how node A can detect it.

7. Repeat the above diagram for a wireless network. Explain why the signal states at node A and node B are no longer identical and why collision detection fails.

8. Why are wireless radios not full duplex from a MAC perspective? What hardware constraint makes simultaneous transmit-and-receive difficult?

4.3. Link-Layer ACK and CSMA-CA

9. Explain how link-layer acknowledgements substitute for collision detection in 802.11. What does the sender infer from the absence of an ACK?

10. Draw the timing diagram for a successful CSMA-CA exchange showing DIFS, Data Transmission, Random Backoff, SIFS, and ACK. Label each interval.

11. What is the advantage of using small control frames (RTS and CTS) in CSMA-CA when data packets are large?

12. Define "virtual carrier sense" and explain how it is implemented by the RTS/CTS mechanism.

4.4. Hidden Terminal and Exposed Terminal Problems

13. Draw the hidden terminal scenario with nodes A, B, and C. Explain why C transmits while A is transmitting to B, and describe the result at B.

14. Explain how the RTS/CTS exchange solves the hidden terminal problem. Trace the sequence of frames and identify which frame causes the hidden node to defer.

15. Draw the exposed terminal scenario with nodes A, B, C, and D. Explain why C unnecessarily defers from transmitting to D.

16. Using the RTS/CTS mechanism, explain how an exposed terminal (C) can determine it is safe to transmit concurrently. Which frame does C hear, and which does it not hear? What does C conclude?

17. State the key insight about the spatial location of collisions and explain why it motivates receiver-based collision control.

4.5. Provisioned MAC Protocols

18. Describe the structure of a TDMA round. In a six-station network where only stations 1, 3, and 4 have data, draw the slot allocation and identify wasted slots.

19. What is an "out-of-band" mechanism in TDMA? Why is it necessary, and what overhead does it introduce?

20. Describe FDMA. Draw a frequency-vs.-time diagram for six stations where only three are active. Identify wasted frequency bands and guard bands.
21. Compare TDMA and FDMA using the following attributes: resource divided, guaranteed access, idle resource waste, coordination overhead, and robustness to fading.
22. A network designer must choose between TDMA and a contention-based protocol for a real-time voice application with 20 users. Which would you recommend and why?

4.6. Spatial Reuse and Carrier Sense Thresholds

23. Explain spatial reuse. Under what conditions should two links transmit concurrently, and under what conditions should they take turns?
24. Draw two network topologies: one where carrier sense enables correct concurrency and one where carrier sense forces unnecessary deferral (exposed terminal). Explain the difference.
25. What is a "thresholded correlation value" in carrier sense? How does it determine whether the medium is occupied?
26. Explain the advantages of spreading a transmission across a wide shared channel (e.g., 100 MHz) compared to assigning each user a narrow dedicated channel.

5. Sample Questions

5.1. Short Questions

1. Define CSMA. What does each letter in the acronym stand for?
2. Why does 802.11 use collision avoidance instead of collision detection?
3. What is "fate sharing of the link" and in which network type does it apply?
4. Name the four frames exchanged in a CSMA-CA transaction.
5. What does the absence of an ACK signal to the sender in 802.11?
6. Define the hidden terminal problem in one sentence.
7. Define the exposed terminal problem in one sentence.
8. In the RTS/CTS mechanism, which frame causes a hidden terminal to defer? Explain briefly.
9. In the RTS/CTS mechanism, why can an exposed terminal safely transmit even after hearing an RTS?
10. What is TDMA? State one advantage and one disadvantage.
11. What is FDMA? State one advantage and one disadvantage.
12. State the key insight about where collisions are spatially located in a wireless network.
13. Define spatial reuse and give one example of when it applies.
14. What is virtual carrier sense, and how is it provided?

5.2. Descriptive Questions

15. Explain why collision detection, which works well in wired Ethernet (CSMA-CD), cannot be directly applied in wireless networks (802.11). Discuss the hidden terminal problem, signal fading, and the hardware challenge of simultaneous transmit-and-receive.

16. Describe the complete RTS/CTS mechanism with a diagram. Explain how it solves the hidden terminal problem, how it partially addresses the exposed terminal problem, and what trade-offs it introduces.

17. Explain the hidden terminal and exposed terminal problems with diagrams. For each problem, state the cause, the consequence, and the solution provided by RTS/CTS.

18. Describe the concept of spatial reuse in wireless MAC. Explain why far-apart links should transmit concurrently and why nearby links must take turns. How does carrier sensing enforce this, and when does it fail?

19. Explain the role of link-layer acknowledgements in 802.11. Why is this mechanism necessary, and what limitation does it have with respect to the hidden terminal problem?

5.3. Analysis and Design Questions

20. A warehouse deploys a WiFi network where forklifts on opposite sides of metal shelving cannot hear each other but both communicate with a central access point. Identify the MAC problem this creates, and explain how the 802.11 RTS/CTS mechanism addresses it.

21. Consider a wireless network with four nodes: A, B, C, and D arranged linearly (A–B–C–D), where each node can hear only its immediate neighbours. Node B wishes to transmit to A, and node C wishes to transmit to D simultaneously. (a) Does this cause a collision? (b) Will plain CSMA prevent C from transmitting? (c) How does RTS/CTS handle this scenario?

22. Evaluate the trade-off between using RTS/CTS for all packets versus using it only for large packets. Under what traffic conditions is the overhead of RTS/CTS justified?

23. Suppose you are designing a MAC protocol for a dense IoT sensor network where devices transmit short packets infrequently. Would you choose a provisioned protocol (TDMA/FDMA) or a contention-based protocol (CSMA-CA)? Justify your choice by considering channel utilization, coordination overhead, latency, and energy consumption.